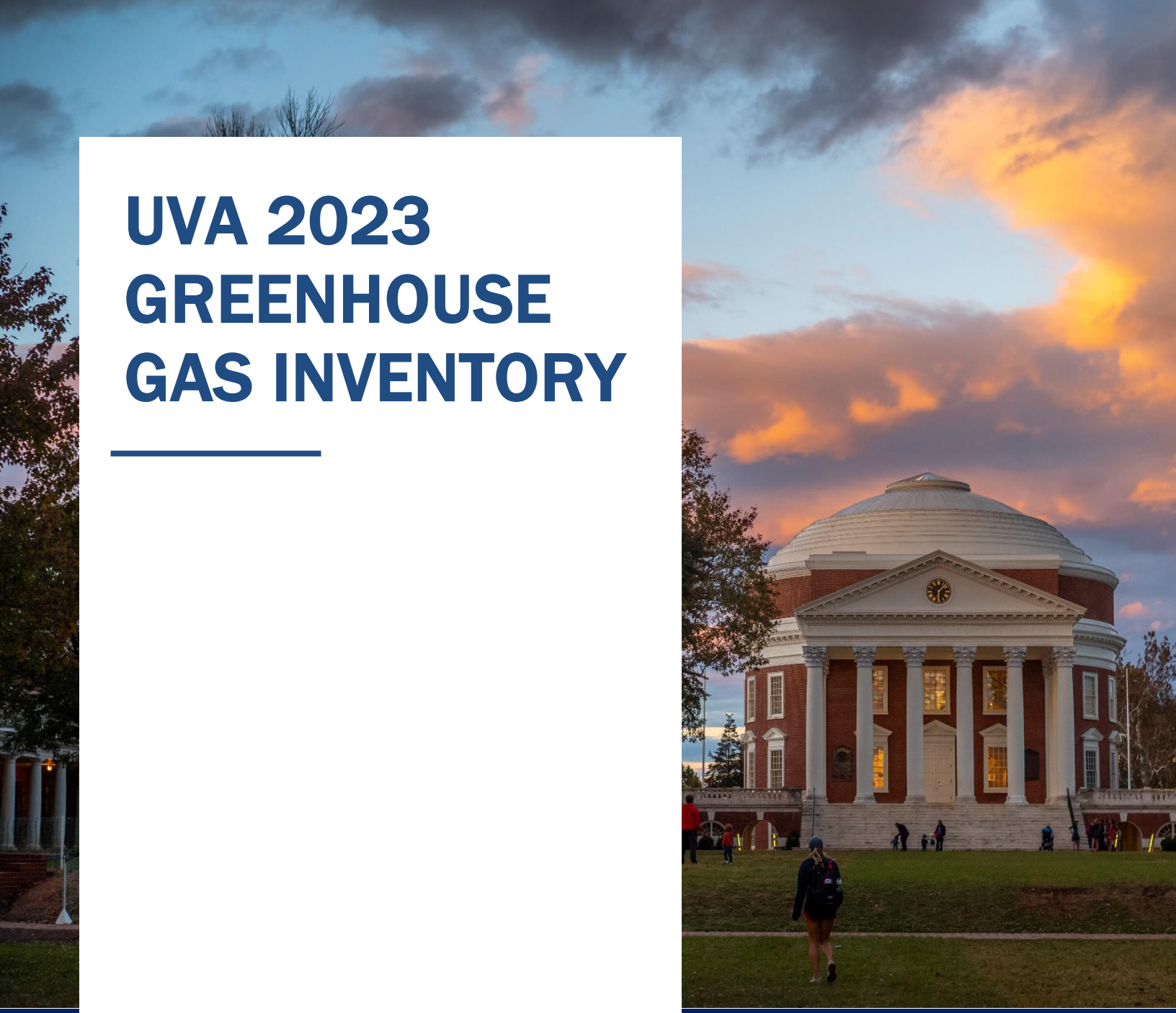
UVA 2023 GREENHOUSE GAS INVENTORY



FALL 2024

UVA OFFICE FOR SUSTAINABILITY

SUSTAINABILITY · UVA
From the Grounds Up

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Executive Summary

In December 2019, the University of Virginia Board of Visitors committed to **be carbon neutral by 2030 and fossil fuel-free by 2050**. In calendar year 2010, the baseline year for emissions analysis, UVA's Scope 1 and 2 GHG emissions footprint was 291,117 metric tons of carbon dioxide equivalent ($MTCO_2e$). In 2023, Scope 1 and 2 emissions decreased to 160,575.6 $MTCO_2e$, resulting in a **44.8% reduction in emissions over 13 years**. This reduction is a significant step towards achieving UVA's climate goals.

UVA's carbon neutrality goal includes greenhouse gas (GHG) emissions related to the operation of the University – commonly referred to as Scope 1 and Scope 2 emissions. This best aligns with the World Resource Institutes' Greenhouse Gas Protocol Corporate Accounting and Reporting Standard's Operational Control Approach, which includes emission sources UVA has the authority to affect through operating policies and initiatives. The University also tracks indirect emissions, known as Scope 3 emissions, from sources such as faculty, staff and student commuting, business air travel, food, and student study abroad, to the extent possible, as part of the University's larger inventory. This inventory tracking will continue to expand and improve as more data continue to be collected and as methodology for Scope 3 calculations become more robust. Each year, UVA's inventory undergoes a detailed third-party data verification review by SIMAP staff to ensure methodologies are consistent and accurate. More information on UVA's climate goals can be found here.

Reduction Strategies

Emissions reductions since 2010 can be attributed to the following initiatives:

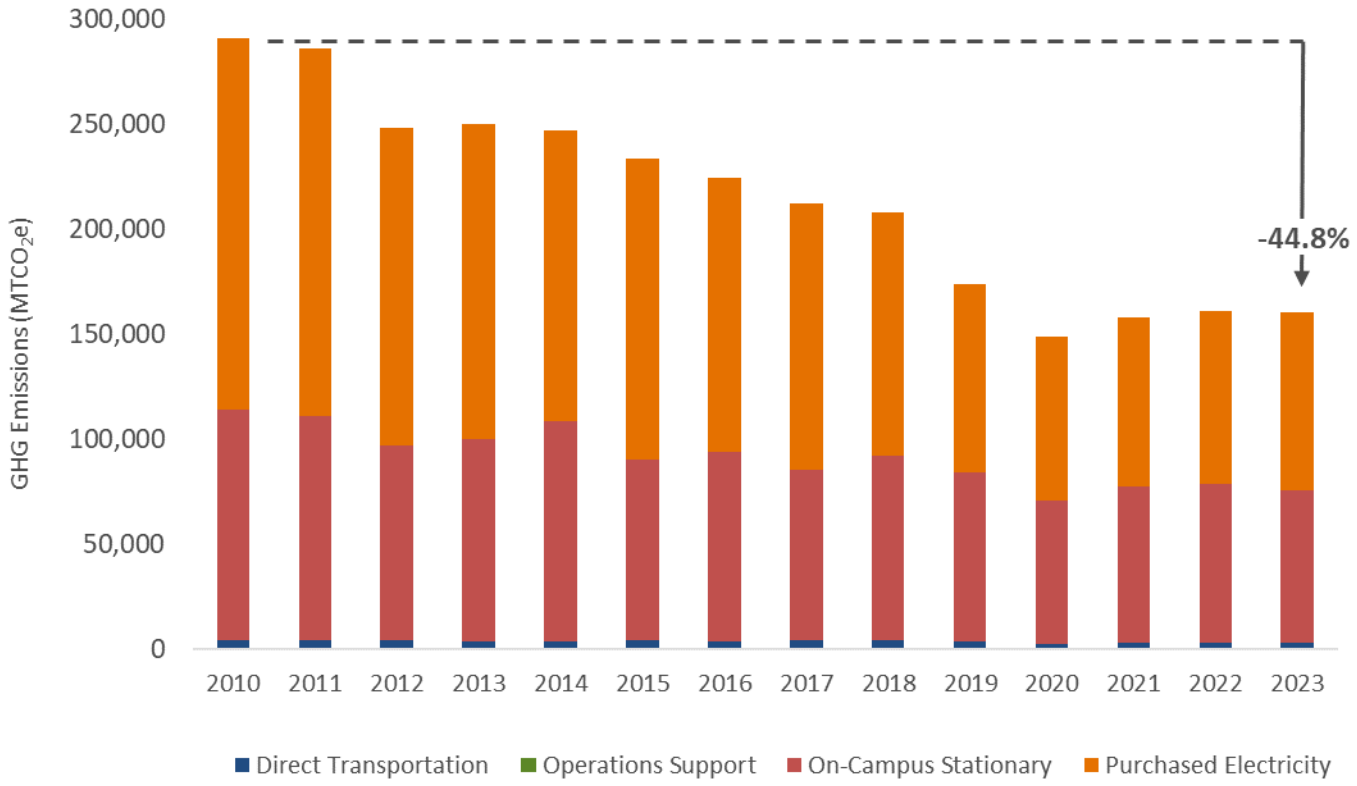
- **Existing Buildings:** Building Efficiency Program, including Sustainable Labs and Sustainable Clinics
- **New Buildings:** UVA Green Building Standards, including energy targets and fossil fuel-free requirements
- **District Energy Upgrades:** Fuel switching in the Main Heat Plant, as well as chilled water, hot water and steam optimization
- **Renewable Energy:** Off-site utility-scale solar and rooftop solar
- **Fleet:** Electric vehicles and buses

GHG Inventory Overview

UVA's Scope 1 and 2 emissions by category is shown in Figure 1 and Table 1:. As this figure illustrates, most of UVA's GHG emissions come from electricity and stationary fuel consumption. The latter is driven by the combustion of natural gas and coal to provide heating for university buildings. Additional sources of GHG emissions come from the University's vehicle fleet (a.k.a. "tailpipe" emissions), fugitive refrigerants and fertilizer. Notably, UVA's GHG emissions reductions have been realized despite continuous growth in the University's operational and physical footprint.

Figure 1: UVA's GHG Emissions by category ($MTCO_2e$), Calendar Years 2010-2023

Table 1: UVA GHG Emissions by category (MTCO2e), Calendar Years 2010-2023



GHG Source	2010	2018	2019	2020	2021	2022	2023
Electricity	176,644	115,710	89,698	77,569	80,633	82,259	84,433
Fuel	109,886	88,229	80,215	68,466	74,292	75,500	72,541
Transportation	4,330	4,067	3,874	2,500	2,887	3,232	3,223
Operations	258	1,682	358	264	391	328	379
Total Net Emissions	**291,117**	**209,687**	**174,144**	**155,095**	**164,747**	**161,319**	**160,576**

***Totals may not sum due to independent rounding.**

Background

In December 2019, the UVA Board of Visitors approved a resolution for **the University to be carbon neutral by 2030 and fossil fuel-free by 2050.** These goals align with UVA's 2030 Great and Good Plan and build upon the successes of the board's 2011 and 2013 sustainability resolutions, the first UVA 2016-2020 Sustainability Plan, and the work of thousands of engaged individuals over the past several years. This report defines UVA's GHG accounting methodology, documents the current footprint, and analyzes the observed emission trends to target areas for further reductions.

Methodology

UVA's GHG Inventory is calculated each calendar year and measured against a 2010 baseline using the web-based Sustainability Indicator Management & Analysis Platform (SIMAP) developed by the University of New Hampshire. The methodologies in this tool align with the World Resource Institutes' Greenhouse Gas Protocol Corporate Accounting and Reporting Standard.

SIMAP calculates emissions from activities that produce carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), refrigerants, and other greenhouse gases. These emissions are converted to metric tons of carbon dioxide equivalent ($MTCO_2e$) using global warming potentials provided by the Intergovernmental Panel on Climate Change's Assessment Reports. Calculation results are reviewed within Facilities Management, by SIMAP reviewers, and then posted publicly on the UVA Sustainability website. In this report, "carbon emissions" and "carbon footprint" refer to the total GHG emissions converted to carbon dioxide emissions equivalent (CO_2e).

Climate Goal and GHG Inventory Boundaries

The 2030 carbon neutrality and 2050 fossil fuel-free goals include all UVA-owned and operated properties in the vicinity of Main Grounds in City of Charlottesville and Albemarle County, including both Academic and Health System (University Medical Center) properties - approximately 570 buildings encompassing over 16 million square feet . The majority of these buildings use thermal energy provided from one of UVA's heating or chilled water plants and electricity via Dominion Energy. UVA's climate goals include all Scope 1 and 2 emissions within this boundary. Per SIMAP recommendations, "de minimis" sources that make up less than 1% of the total scope 1 and 2 footprint may be excluded from this calculation if necessary. More information can be found on UVA's Climate Goal FAQ webpage.

As additional properties contiguous to Main Grounds are acquired by UVA, and as new buildings are constructed, they are added to the inventory (if UVA has operational control) and factored in as growth. Remote buildings under UVA's operational control are included when utility use is known. Additional buildings will be added as/if this information becomes available back to the baseline year. Non-contiguous UVA Real Estate Foundation land, as well as University Physicians Group, College at Wise, leased, and satellite properties are not currently included in the goal because they are either outside of UVA's operational control or utility data is not available.

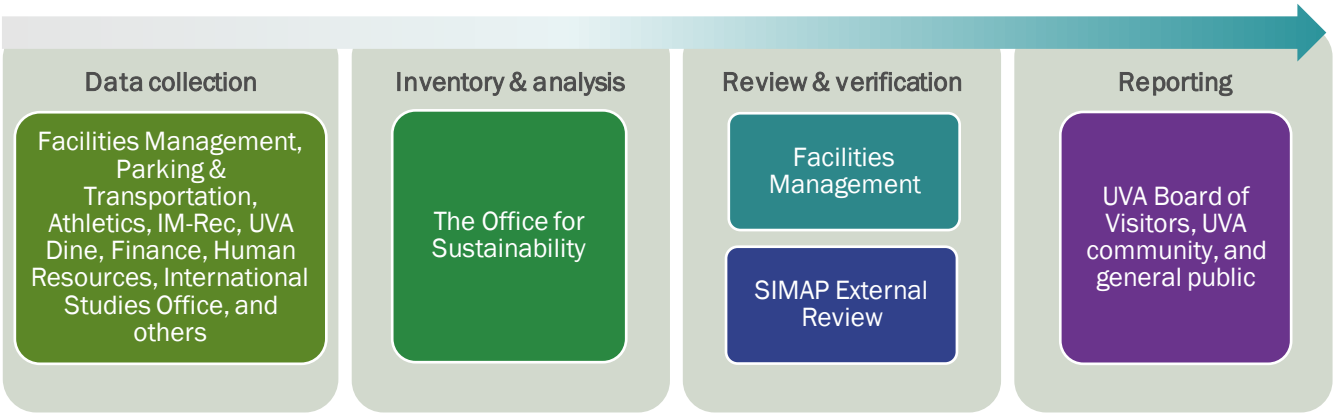
While Scope 3 emissions (e.g., indirect emissions from commuting and food) is not included in the 2030 carbon neutrality goal, they are included in UVA's annual GHG inventory as UVA seeks to actively reduce these emissions. While not directly included in the GHG emissions reduction goal, the University will continue to track, manage, and mitigate indirect Scope 3 emissions where possible. As better data on additional sources of Scope 3 emissions becomes available over time and methodology becomes more robust and accurate, these sources will be incorporated into the University's GHG footprint. For example, Scope 3 emissions associated with university-affiliated air travel (e.g., study abroad travel) began to be included in the 2021 GHG inventory for the first time. More information is provided in the *Scope 3 Sources* section below.

		Greenhouse Gas Inventory	Carbon Neutral Goal	Fossil Fuel Free Goal
Scope 1 & 2	Refrigerants, Fertilizer, & Chemicals	Yes	Yes	N/A
	Fuel - Transport	Yes	Yes	Yes
	Fuel - Buildings & District Utilities	Yes	Yes	Yes
	Purchased Electricity	Yes	Yes	Yes
Scope 3	Commuting	Yes	No	No
	Food	Yes	No	No
	Travel (Business & Study Abroad)	Yes	No	No
	Fuel and Energy-Related Activities (FERA)	Yes	No	No

Data Collection

Data input into SIMAP is derived from a variety of University personnel and sources. Fuel and electricity consumption are metered, and records are retained by Facilities Management. Direct transportation fuel consumption is maintained by Parking and Transportation. Records of refrigerant use are sourced from a variety of locations, with Facilities Management and UVA Dine being the primary contributors. Fertilizers are used and logged by Landscaping (Facilities Management), Athletics, and Intramural-Recreational Sports. Data for estimating Scope 3 emissions are provided by numerous UVA schools and business units, including Procurement and Supplier Diversity Services, the University of Virginia Health System, UVA Dine, Darden School of Business, the International Studies Office (ISO), and others. Figure 2 portrays UVA's institutional arrangements regarding the GHG Inventory process.

Figure 2: UVA GHG Inventory Institutional Arrangements

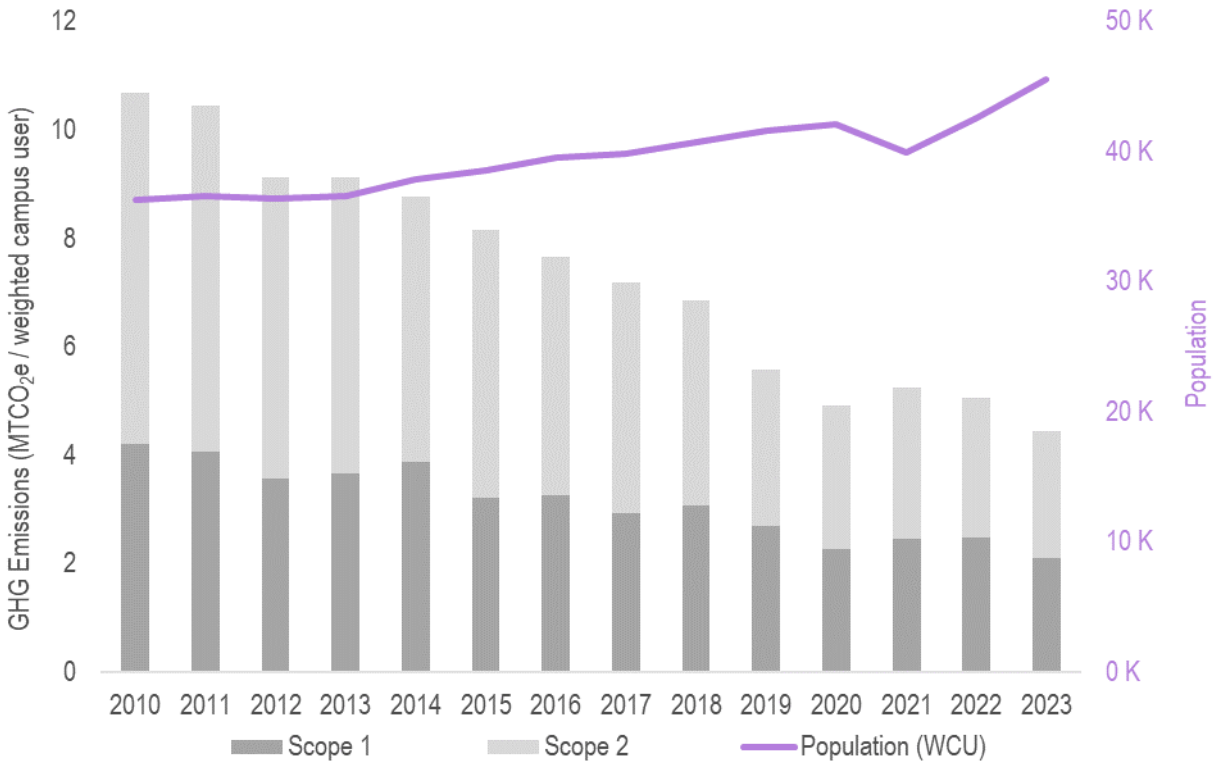


Addressing growth

Expansion of the University's offerings has resulted in growth in both population and building area. Since 2010, the population has increased by 6,047 students, faculty, and staff (16.6% increase). Approximately 3.3 million square feet (22.1% increase) of building space has been added to UVA's footprint and included within the boundaries defined for UVA's GHG inventory. Despite growth, UVA has reduced its normalized, Scope 1 and 2 emissions both per weighted campus user* and by square foot. In 2010, UVA reported 10.7 MTCO2e per person and 19.2 MTCO2e per thousand square feet. In 2022, these numbers decreased to 4.45 MTCO2e per person and 8.98 MTCO2e per thousand square feet, respectively. The University's commitment to achieving its carbon neutrality and fossil fuel-free goals, considering its continuing expansion, ensures integration of sustainability in renovations and new construction to enhance the University's operations and building portfolio.

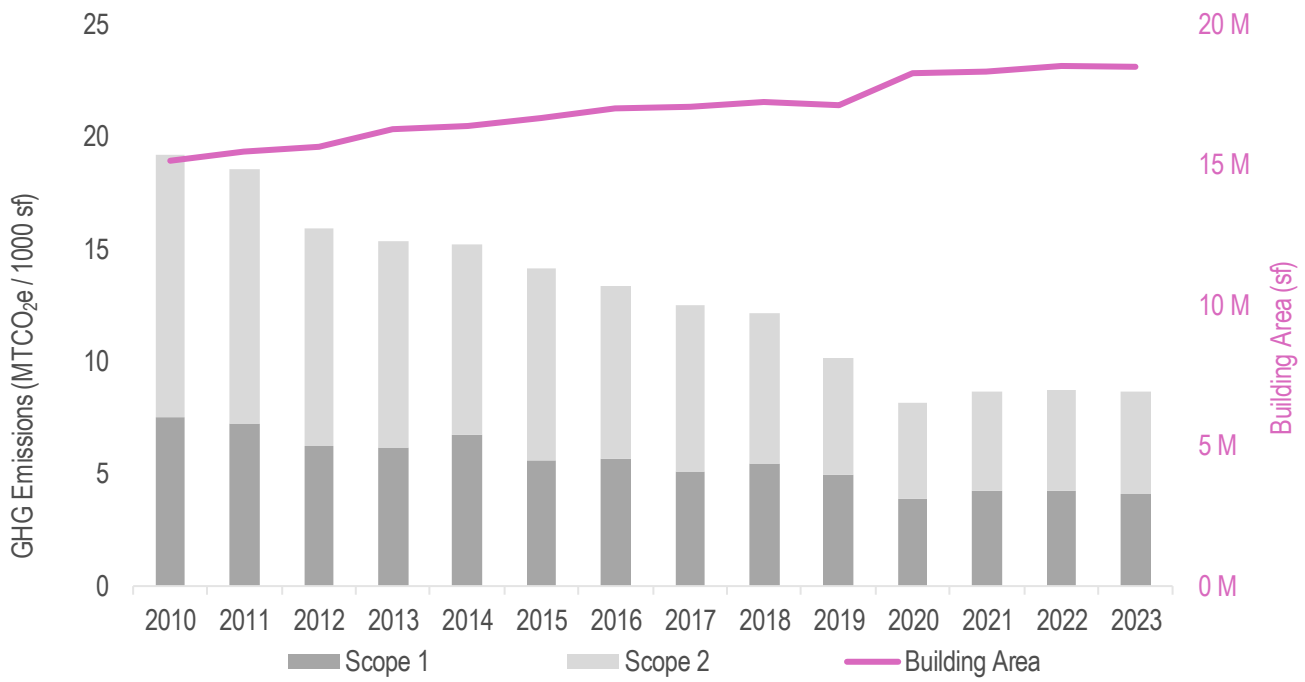
Figure 3 and 4 report UVA's normalized 2023 GHG emissions by scope.

Figure 3: UVA Normalized GHG Emissions by scope ($MTCO_2e$ / weighted campus users), 2010-2023



**Weighted campus user is [based on AASHE STARs 2.1 definition]*

Figure 4: UVA Normalized GHG Emissions by scope ($MTCO_2e$ / 1,000 gross square feet), 2010-2023



Emissions by Scope

When comparing the University's emissions by scope, several trends emerge. As Figure 5 illustrates, Scope 1 and 2 emissions have decreased over time. Scope 3 emissions exhibit a less consistent pattern over time, but this can be attributed to additional data becoming available and additional sources being added to the Scope 3 inventory over time. Scope 1 emissions are those produced by the combustion of fossil fuels in buildings, heating plants, and University fleet vehicles, as well as smaller categories like refrigerants and fertilizer. The decrease in Scope 1 emissions is mainly driven by district energy system optimization, which includes fuel switching, equipment upgrades, and connecting more facilities to UVA's centralized heating and cooling systems. Scope 2 emissions are associated with purchased electricity and have also decreased due to energy efficiency upgrades in buildings and energy systems over time. Electricity-related emissions have dropped even more as the regional grid has become less carbon intensive and the University has procured renewable electricity. The variation in Scope 3 emissions represents the increase in total sources of emissions over time, adding more categories of data to the calculations, as well as a time lag in the availability of data across a decentralized institution. Estimates for University-affiliated air travel emissions, for example, were only added to the inventory in 2021. Efforts to populate historical data back to 2010 are ongoing. Food-related emissions are available through 2022 (with 2023 emissions inventory still in the works). More information on changing emissions factors can be found in section 2 of the appendix.

Scope 2 emissions, from purchased electricity, narrowly constitute the largest contributor to UVA's GHG footprint (excluding scope 3 emissions outside of the University's goal). In 2023, Scope 2

emissions accounted for 52.6% of total emissions. Scope 1 emissions contributed 47.4%. Figure 5 and Table 2 report UVA's 2022 GHG emissions by scope.

Figure 5: UVA GHG Emissions by scope (MTCO2e), 2010-2023

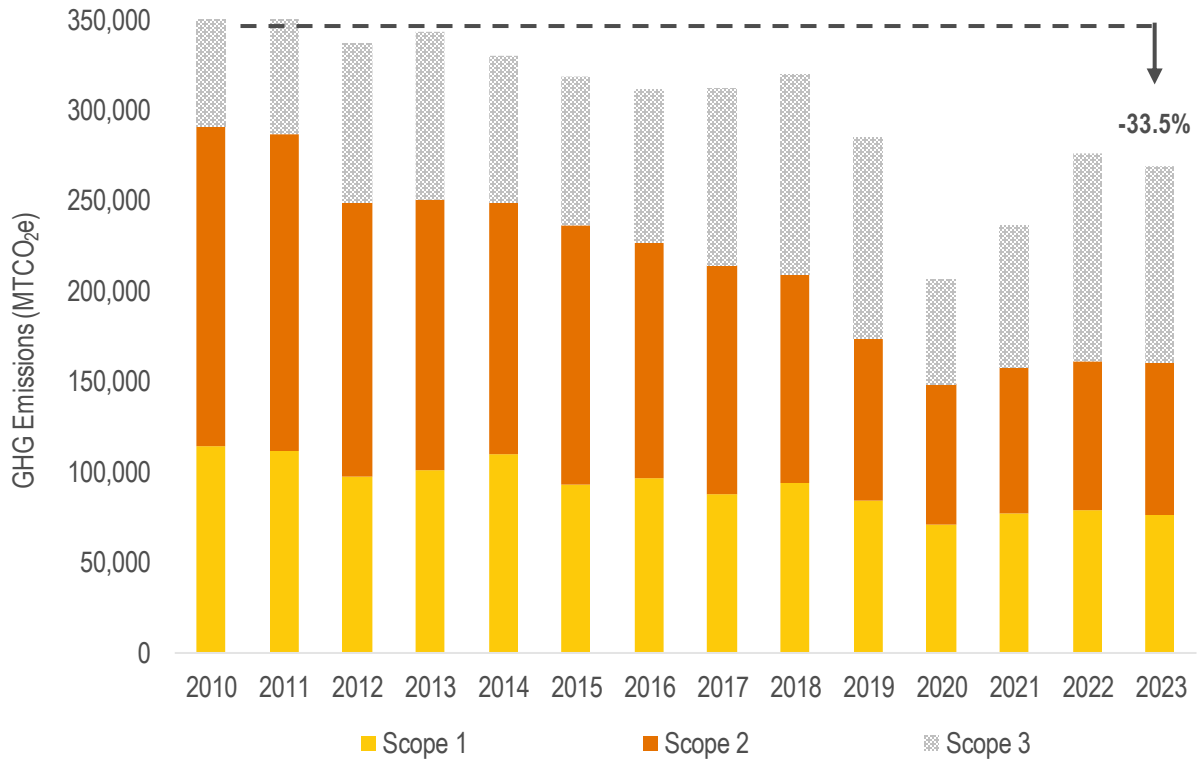


Table 2: UVA GHG Emissions by scope (MTCO2e), 2010-2023

GHG Scope	2010	2018	2019	2020	2021	2022	2023
Scope 1	114,473	93,977	84,446	71,230	77,570	79,060	76,143
Scope 2	176,644	115,710	89,698	77,569	80,633	82,259	84,433
Scope 3[a]	77,284	110,355	111,247	58,100	78,630	103,954	109,168
Total Gross Emissions (excluding Scope 3)	**291,117**	**209,687**	**174,144**	**148,799.6**	**158,203**	**161,319**	**160,576**

*Totals may not sum due to independent rounding.

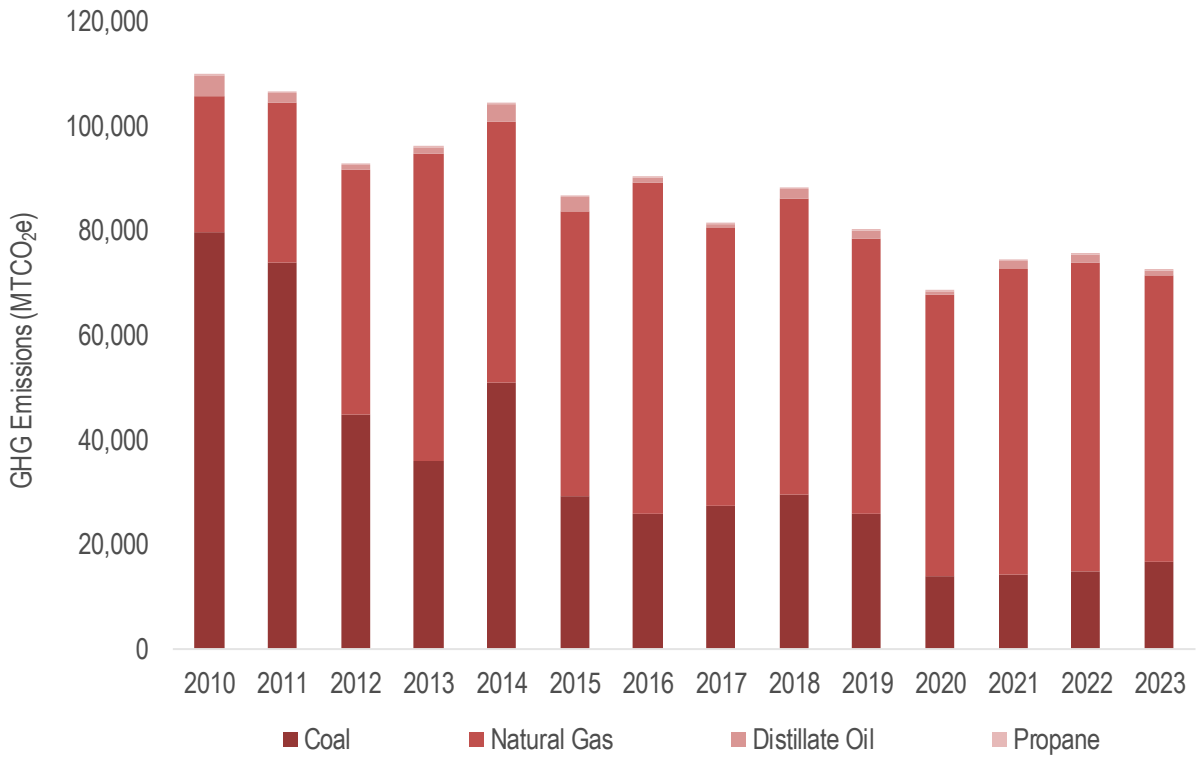
**Refer to Appendix A Table 4 for a complete Scope 1&2 emission data set from 2010 to 2023.

a. UVA tracks Scope 3 emissions, however they are not considered part of UVA's carbon neutral and fossil fuel-free goals

Fuel (On-Grounds Stationary Sources)

Most of UVA's Scope 1 emissions stem from on-Grounds stationary fuels used for heating. Additional uses of stationary fuels include emergency and temporary generators. These sources include natural gas, coal, distillate oil, and propane gas. In 2023, these sources accounted for 76,143 MTCO2e, or 45.2% of UVA's emissions. This total was an 3.9% decrease from 2022 and a 34.0% decrease from 2010 levels.

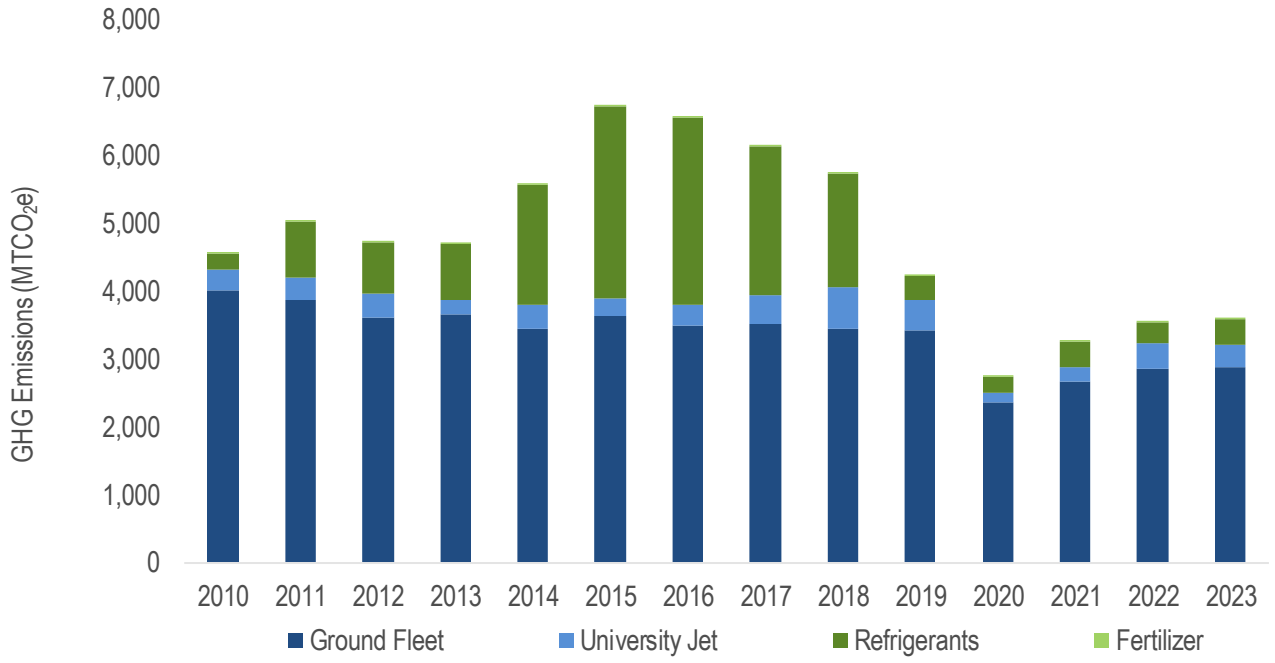
Figure 6: UVA On-Grounds Stationary Combustion Emissions (MTCO2e) Trends, 2010-2023



Transportation & Operations

The UVA ground-vehicle fleet includes the University Transit System buses, cars, and maintenance vehicles that are owned and operated by the university. These vehicles typically consume gasoline, diesel, or biodiesel fuel. The University operates twenty-three electric vehicles whose emissions contributions are included with Scope 2 electricity-related emissions, as they do not contribute direct Scope 1 emissions. UVA vehicles and the University's jet accounted for 3,223.1 $MTCO_2e$, in 2023. Together, these sources equal approximately 2.0% of total Scope 1 and 2 emissions. Additionally, UVA calculates the emissions generated from the release of refrigerants and other select chemicals into the atmosphere as well as off-gassing from fertilizers. In 2023, these sources represent 364.2 and 14.5$MTCO_2e$, respectively, or less than 1% of Scope 1 and 2 emissions.

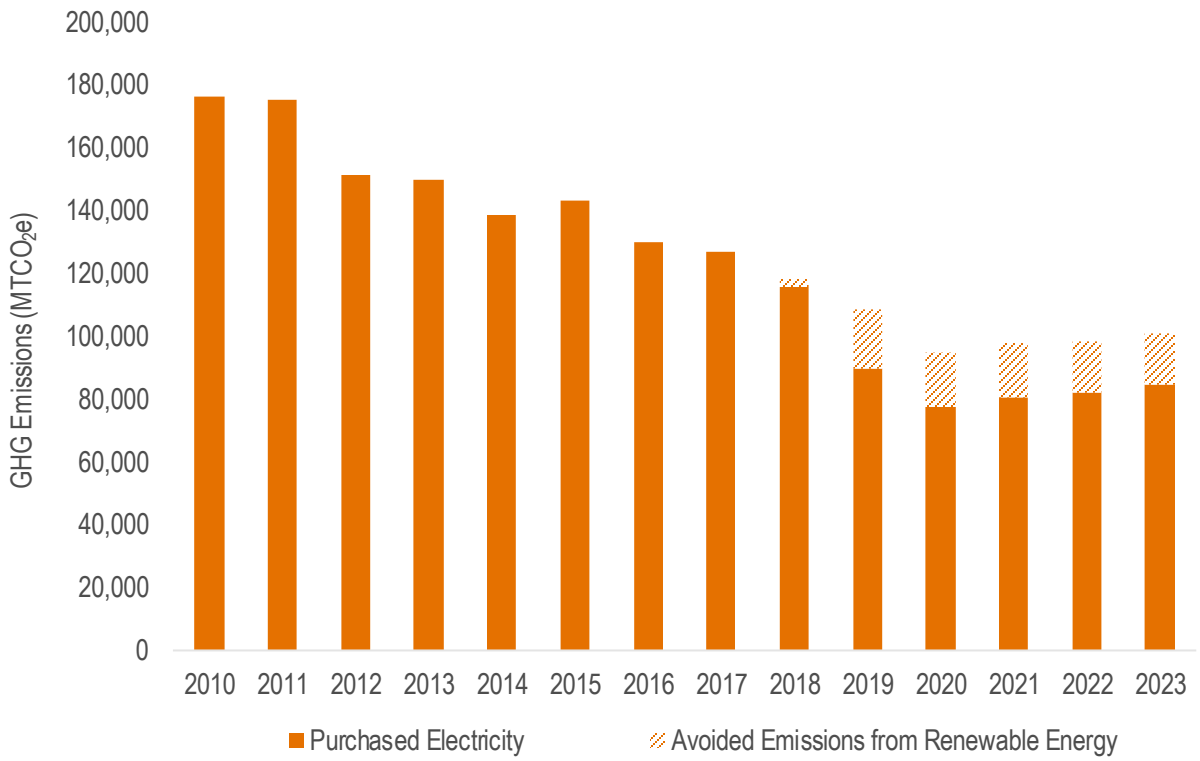
Figure 7: UVA Transportation and Operations Emissions ($MTCO_2e$) Trends, 2010-2023



Electricity

Electricity consumption, used primarily for cooling and electrical services such as lighting and equipment in buildings, continues to be the largest source of GHG emissions at UVA. In 2023, purchased electricity accounted for 84,432.5 MTCO2e, or 52.6% of Scope 1 and 2 emissions. This was a 2.6% increase from 2022 and a 52.2% decrease from 2010. Additionally, through a power purchase agreement with Dominion Virginia Power, UVA is generating approximately 60,000 megawatt hours of solar renewable electricity annually. This electricity is generated from two off-site, utility-scale solar fields, known as Hollyfield and Puller. Together, these installations help UVA avoid approximately 20,000 $MTCO_2e$ of emissions each year.

Figure 8: UVA Electricity GHG Emissions by Source ($MTCO_2e$), 2010-2023



Scope 3 Categories

In addition to the categories included in UVA's GHG reduction goal, the University collects data on emissions for several indirect sources including student, faculty and staff commuting, directly financed air travel, study abroad air travel, waste, wastewater, food, electricity transmission and distribution (T&D) losses, and other fuel- and energy-related activities (FERA). UVA will continue to calculate additional emissions sources as data becomes available. Additional Scope 3 emissions categories may not be included due to their de minimis contribution. Thus, Scope 3 emissions numbers will likely increase as more data is added. Regardless, UVA is committed to implement strategies to reduce these emissions to the extent possible.

Much of UVA's commuting emissions is captured in the City of Charlottesville and Albemarle County's greenhouse gas inventories because of the way the Virginia Department of Transportation collects that data, based on vehicle miles traveled. For UVA's inventory, commuting emissions are estimated from the approximate miles, frequency and mode of transportation students, faculty, and staff utilize to travel to and from the University each day. This information is collected through a University-wide survey that is conducted every three years. The survey was postponed in through 2022 due to the dramatic change in typical commuting practices in response to the COVID-19 pandemic. Following guidelines proposed by SIMAP and university policies, estimates were made based on university populations through 2020 and 2022 to

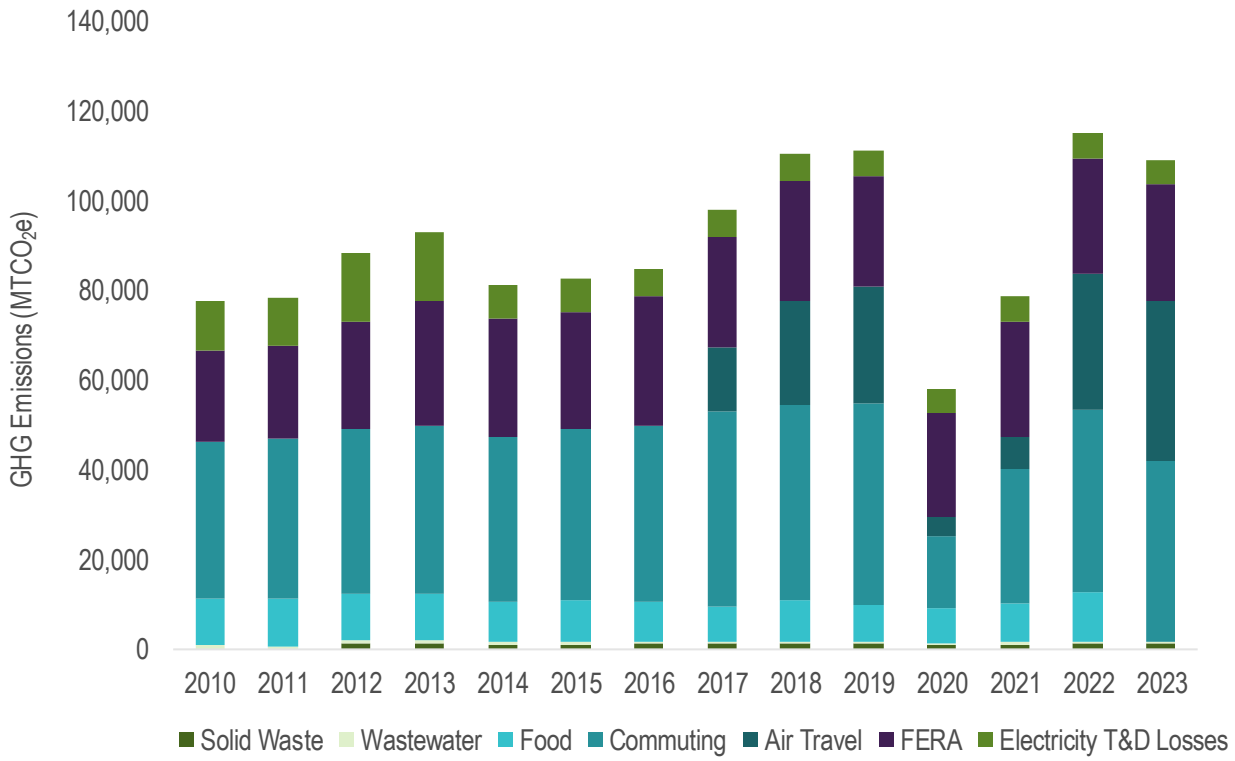
determine a better estimate for community during the COVID-19 pandemic. The most recent commuting survey was conducted in 2023 and the new data are included.

Waste-related data are collected and reported by Facilities Management. Wastewater emissions are estimated based on monthly sewer charges to the University.

Food data is collected as part of the Nitrogen goal tracking process based on data from dining halls, cafes, and restaurants on Grounds, as well as various catering services. Emissions factors for this data are calculated based on factors in SIMAP. The food emission inventory has a 1-year delay due to the extra time needed to gather and validate such large amount of data.

Upstream energy emissions refer to GHG emissions associated with energy supply chains that are otherwise not included in Scopes 1 and 2. Examples include energy transmission and distribution losses (e.g., power lines, natural gas pipelines). Purchased electricity's transmission and distribution (T&D) losses are calculated as a percentage of total electricity consumption based on regional estimates provided by the U.S. Environmental Protection Agency. Upstream emissions associated with fossil fuel consumption (natural gas, coal, and fuel oil) are associated with the extraction, refining, storage, and transport of each fuel.

Figure 9: UVA Scope 3 Emissions ($MTCO_2e$) Trends, 2010-2023



It is important to note there are Scope 3 GHG emission categories that are not included or recorded completely in UVA's GHG Inventory such as holiday travel. The study abroad air travel includes undergraduate programs from International Studies Offices and graduate programs data from Darden School of Business, UVA Law School, McIntire School of Commerce, and the School

of Medicine. These emission categories have not been included or recorded completely because University-wide reporting methods do not currently exist in the appropriate capacity to generate consolidated and complete information.

Appendix A: Emissions and energy trends

Table 3: UVA GHG Emissions by scope (MTCO2e), 2010-2023 Complete Data

	2010	2011	2012	2013	2014	2015	2016
Scope 1	114,473	111,533	97,474	100,867	110,109	93,295	96,845
Scope 2	176,644	175,506	151,650	149,829	138,886	143,030	130,274
Total Gross Emissions	**291,117**	**287,038**	**249,124**	**250,696**	**248,995**	**236,325**	**227,119**

(Continued)

	2017	2018	2019	2020	2021	2022	2023
Scope 1	87,442	93,977	84,446	71,230	77,570	79,060	76,143
Scope 2	126,882	115,710	89,698	77,569	80,633	82,259	84,432
Total Gross Emissions	**214,324**	**209,687**	**174,144**	**148,800**	**158,203**	**161,319**	**160,576**

*Totals may not sum due to independent rounding

Table 4: UVA Electricity Emissions (MTCO2e) Trends, 2010-2022

GHG Source	2010	2018	2019	2020	2021	2022	2023
Purchased Electricity	176,644	115,710	89,698	77,569	80,633	82,259	84,433
Avoided Emissions from Solar	-	(2,803)	(18,891)	(17,251)	(17,190)	(16,030)	(16,694)

*Totals may not sum due to independent rounding.

Table 5: UVA On-Grounds Stationary Combustion Emissions (MTCO2e) Trends, 2010-2023

GHG Source	2010	2018	2019	2020	2021	2022	2023
Coal	79,736	29,614	25,831	13,897	14,249	14,748	16,566
Natural Gas	26,037	56,710	52,845	53,766	58,608	59,060	52,053
Distillate Oil	3,969	1,697	1,379	738	1,309	1,581	832
Propane	145	208	161	66	126	111	90
Total Gross Emissions	**109,886**	**88,229**	**80,215**	**68,466**	**74,292**	**75,500**	**72,541**

*Totals may not sum due to independent rounding.

Table 6: UVA Transportation Emissions (MTCO2e) Trends, 2010-2023

GHG Source	2010	2018	2019	2020	2021	2022	2023
B100	272	92	94	58	64	68	65
Diesel	2,107	1,874	1,902	1,175	1,295	1,373	1,317
Gasoline	1,647	1,498	1,434	1,146	1,324	1,418	1,504
Jet Fuel	304	603	444	122	204	374	338
Total Gross Emissions	**4,330**	**4,067**	**3,874**	**2,500**	**2,887**	**3,232**	**3,223**

*Totals may not sum due to independent rounding.

Table 7: UVA Operations Support Emissions (MTCO2e) Trends, 2010-2023

GHG Source	2010	2018	2019	2020	2021	2022	2023
Refrigerants & Chemicals	240	1,666	347	254	379	317	364
Fertilizer	18	15	11	11	12	12	15
Total Gross Emissions	258	1,682	358	265	391	328	379

*Totals may not sum due to independent rounding.

Table 8: UVA Scope 3 Emissions (MTCO2e) Trends, 2010-2023

GHG Source	2010	2018	2019	2020	2021	2022	2023
Faculty Commuting	5,155	5,688	5,789	2,326	4,235	5,353	5,276
Staff Commuting	27,461	37,179	38,380	13,877	25,558	34,923	34,462
Student Commuting	2,476	960	811	89	224	297	295
Directly Financed Air Travel	0.0	23,114	26,131	4,039	6,687	22,843	27,973
Study Abroad Air Travel	0	0	0	0	597	7,372	7,669
Solid Waste	(237)	1,310	1,252	1,062	1,144	1,218	1,287
Wastewater	894	520	521	414	452	468	580
Food	10,401	8,975	8,060	7,546	8,586	11,093	N/A
FERA	20,217	26,441	24,469	23,441	25,673	25,978	25,979
T&D Losses	10,918	6,083	5,836	5,307	5,475	5,502	5,647
Total Gross Emissions	**77,284**	**110,355**	**111,247**	**58,100**	**78,630**	**103,954**	**109,168**

*Totals may not sum due to independent rounding.

Appendix B: Additional Information

Emissions Factors

Year to year emissions factors for several sectors may change due to new data sets becoming available or new methodologies are recommended. For example, electricity emissions factors for greenhouse gases change on an annual basis with an expected two-year lag for the eGRID to publish updates. We expect and account for updates to the eGRID emission factors on a yearly basis. Other instances of emissions factor changes are when new methodologies are recommended by the GHG protocol to better estimate emissions. These happen on a more infrequent basis and are often alerted to our team through announcements from the SIMAP platform. Updating our values – which in turn means updating numbers in prior year's reports to reflect new methodologies allows us to make the best estimate of UVA's emissions. This the primary reason a prior year's total emissions may change in subsequent reports.

Additional Resources

This section provides external links to resources listed within the report.

1. Sustainable Indicators Management and Analysis Platform (SIMAP): Platform used to tabulate GHG footprint on an annual basis with information provided by stakeholders.

2. Charlottesville City Climate Action Plan: Explains what is included in the Charlottesville City GHG inventory and what actions are being taken to reduce emissions.
3. Albemarle County Climate Action Plan: Explains what is included in the Albemarle Climate GHG inventory and what actions are being taken to reduce emissions.